INTRODUCTION

Science is humanity's ongoing attempt to understand how the Universe works. Whether in the form of meticulous research, ingenious insight, or unexpected discoveries, the search for truth continues to challenge, often leading to more questions than answers.

Throughout history, humans have always been driven to understand the workings of the world. In their quest for knowledge, philosophers in ancient Greece were the first to try to explain what they observed. However, many of their ideas were inaccurate, as their philosophical method lacked any experimental evidence. Thales of Miletus in the 6th century BCE, for example, proposed that water is the primary substance of the cosmos, as he had realized that water is essential for life and that it exists on land, sea, and in the air. Two centuries later, Aristotle wrote widely on scientific subjects, from physics and biology to astronomy, laying the foundations for much of the work that has followed. However, he also made fundamental errors, such as arguing that heavier objects fall faster than lighter ones, because he relied on thought and argument rather than empirical proof.

Despite being some distance from a reliable scientific method—in which a hypothesis is proposed, systematically tested, and evaluated on the strength of the results—the early thinkers made vital findings. Halley's comet was observed in China in 240 BCE, and some 300 years later, Zhang Heng explained eclipses and drew up an extensive catalog of stars.

Scientific dawn

While European progress in science stalled during the Middle Ages, the shift of knowledge to the Islamic world inspired rapid advances in scientific thinking. The move of most of the important writings from Greece and India to The House of Wisdom—the library of the Abbasid caliphate in Baghdad—in the 8th century CE nurtured scholars. The scholars included the mathematician al-Khwarizmi, whose works included trigonometry, algebra, and astronomy, and Alhazen, an innovator in the field of optics, whose studies of dissected bulls' eyes were the first scientific experiments.

Progression of ideas

A mark of a true scientist is the ability to evaluate and revise previously held truths. Today, scientists understand that